

BROWNSVALE AGCM, AB, Canada

WMO: 712830

Lat: **56.1178N**Lon: **117.8864W**Elev: **2454**StdP: **13.44**Time Zone: **-7.00 (NAM)**Period: **08-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-25.8	-20.4	-32.1	1.0	-25.6	-26.6	1.4	-20.2	26.0	33.4	22.8	29.9	3.7	20	0.673

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.5	81.0	60.7	78.0	59.2	75.3	58.2	63.5	75.6	61.9	74.0	60.3	71.7	8.0	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.0	81.7	68.2	57.2	76.5	66.4	55.5	71.8	64.6	30.1	76.2	28.9	73.8	27.7	71.6	70.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.4	18.9	16.7	DB	-33.1	86.0	5.2	2.1	-36.8	87.5	-39.9	88.8	-42.8	89.9	-46.6	91.5	(4)
(5)			WB	-33.3	67.0	5.1	2.4	-37.0	68.7	-40.0	70.1	-42.9	71.4	-46.6	73.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	35.6	11.2	11.2	21.1	37.2	50.9	57.7	61.1	59.2	50.4	37.6	19.6	8.7
	DBStd	22.24	15.83	15.01	14.05	9.01	7.88	5.64	4.66	5.86	7.93	8.82	12.72	14.36
	HDD50	6349	1202	1086	896	392	85	8	1	5	90	392	912	1280
	HDD65	10755	1667	1506	1361	833	438	225	137	193	440	848	1362	1745
	CDD50	1107	0	0	0	9	113	240	345	290	101	9	0	0
	CDD65	38	0	0	0	0	1	7	16	13	1	0	0	0
	CDH74	862	0	0	0	6	76	164	309	243	64	0	0	0
	CDH80	130	0	0	0	0	12	27	37	48	6	0	0	0
	WSAvg	7.2	7.5	7.0	7.5	8.2	8.4	7.2	6.6	6.4	6.8	7.1	6.4	6.9
	PrecAvg	16.3	0.9	0.6	0.7	0.7	1.6	2.9	2.8	1.9	1.4	0.9	0.9	0.8
(15) Precipitation	PrecMax	21.3	2.2	1.4	2.0	1.9	3.6	5.3	5.1	4.9	4.1	2.4	2.1	2.8
	PrecMin	9.9	0.2	0.2	0.1	0.0	0.6	0.7	1.1	0.7	0.0	0.2	0.2	0.1
	PrecStd	3.6	0.6	0.3	0.4	0.4	0.7	1.4	1.0	1.1	0.8	0.5	0.6	0.6
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.0	42.2	50.5	71.7	81.1	83.8	83.7	85.3	79.8	69.3	47.9	41.5
		MCWB	36.4	36.4	40.7	50.3	55.4	61.4	62.5	61.8	59.4	53.0	40.9	35.8
	2%	DB	40.7	39.5	46.3	62.2	75.2	78.1	80.4	80.1	75.4	59.7	42.5	38.0
		MCWB	35.9	34.6	38.4	45.2	53.0	58.5	61.5	60.9	57.5	48.0	37.2	33.2
	5%	DB	38.5	36.5	42.3	55.8	70.6	75.0	77.5	76.2	69.8	55.0	39.7	33.9
		MCWB	34.5	32.5	35.6	42.1	50.8	57.0	60.0	59.7	54.8	44.7	35.1	30.1
	10%	DB	34.9	32.5	39.3	51.3	67.1	71.6	74.7	72.9	64.9	50.8	36.1	29.5
		MCWB	31.6	29.3	33.5	40.2	49.4	55.9	59.3	58.7	52.6	42.3	32.3	26.6
	0.4%	WB	37.5	37.1	41.7	51.9	57.4	63.9	67.2	65.1	61.7	54.1	41.8	36.2
		MCDB	41.6	41.4	49.3	71.7	74.9	78.9	79.8	77.4	76.3	66.3	46.8	40.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	36.0	35.0	38.7	46.5	54.7	61.4	63.9	63.1	58.6	49.3	37.9	33.1
		MCDB	40.0	38.7	46.0	61.4	69.6	74.3	74.5	75.5	71.7	58.3	42.6	37.3
	5%	WB	34.6	32.6	36.0	42.8	52.5	59.0	62.2	61.6	56.1	45.5	35.0	30.1
		MCDB	38.5	36.5	41.8	54.0	67.2	70.8	73.7	73.4	67.9	53.0	38.9	33.8
	10%	WB	31.5	29.3	33.9	40.5	50.7	57.1	60.6	59.7	53.8	43.2	32.7	26.4
		MCDB	35.1	32.5	38.9	50.3	64.7	68.8	71.7	70.4	64.2	50.0	35.7	29.2
(35) Mean Daily Temperature Range	5% DB	MDBR	15.5	17.1	18.7	18.9	23.5	21.2	22.5	22.8	22.3	16.4	14.4	15.2
		MCDBR	14.0	17.2	20.5	26.6	30.0	28.4	28.5	28.6	30.7	23.4	16.6	17.4
		MCWBR	11.1	13.4	14.0	14.0	13.5	12.5	12.6	12.5	14.6	13.8	12.5	14.6
	5% WB	MCDBR	14.2	16.7	19.5	25.2	27.0	24.1	25.4	25.5	29.0	22.4	16.5	17.6
		MCWBR	11.6	13.3	13.6	13.9	12.6	11.6	12.7	12.7	14.5	13.7	12.9	15.0
(40) Clear-Sky Solar Irradiance	taub		0.245	0.242	0.287	0.318	0.333	0.343	0.344	0.341	0.303	0.277	0.236	0.220
	taud		2.235	2.351	2.310	2.392	2.419	2.410	2.423	2.418	2.536	2.561	2.417	2.223
	Ebn at Noon		216	272	281	287	288	285	283	276	275	258	227	198
	Edh at Noon		16	22	30	32	33	34	33	31	24	18	14	13
(44) All-Sky Solar Radiation	RadAvg		224	532	1020	1488	1776	1904	1853	1505	1030	548	242	153
	RadStd		22	45	82	89	130	153	116	128	84	67	21	12

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (20 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+103	N/A

Nomenclature: See separate page